more electrons being hit and knocked clear off the surface. But notice that the leftover energy that each of these electrons has after ripping free of the surface depends solely on the energy of the photon that hits it—and this is determined by the frequency of the light beam, not its total intensity. Just as children leave the basement with 15 cents no matter how many dollar bills are thrown down, each electron leaves the surface with the same energy—and hence the same speed—regardless of the total intensity of the impinging light. More total money simply means more children can leave; more total energy in the light beam simply means more electrons are knocked free. If we want children to leave the basement with more money, we must increase the monetary denomination tossed down; if we want electrons to leave the surface with greater speed, we must increase the frequency of the impinging light beam—that is, we must increase the energy denomination of the photons we shine on the metallic surface.

This is precisely in accord with the experimental data. The frequency of the light (its color) determines the speed of the ejected electrons; the total intensity of the light determines the number of ejected electrons. And so Einstein showed that Planck's guess of lumpy energy actually reflects a fundamental feature of electromagnetic waves: They are composed of particles—photons—that are little bundles, or quanta, of light. The lumpiness of the energy embodied by such waves is due to their being composed of lumps.

Einstein's insight represented great progress. But, as we shall now see, the story is not as tidy as it might appear.

Is It a Wave or Is It a Particle?

Everyone knows that water—and hence water waves—are composed of a huge number of water molecules. So is it really surprising that light waves are also composed of a huge number of particles, namely photons? It is. But the surprise is in the details. You see, more than three hundred years ago Newton proclaimed that light consisted of a stream of particles, so the idea is not exactly new. However, some of Newton's colleagues, most notably the Dutch physicist Christian Huygens, disagreed with him and argued that light is a wave. The debate raged but ultimately experiments carried out by the English physicist Thomas Young in the early 1800s showed that Newton was wrong.